# Punya Mittal

Software Engineer — AI/ML — Backend Systems

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## Professional Summary

Software Engineer and B.Tech. CS (AI) student with experience in building backend systems, REST APIs, data pipelines, and machine learning systems. Skilled in Python, Java, Node.js, SQL, machine learning, and distributed systems, with hands-on experience owning end-to-end engineering workflows.

## Education

**Vellore Institute of Technology (VIT), Chennai** 2024 – 2028

*B.Tech. Computer Science and Engineering (Artificial Intelligence)* *CGPA: 8.55 / 10*

## Technical Skills

- **Languages:** Python, Java, JavaScript, TypeScript, SQL
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design, Distributed Systems
- **Backend/Web:** Node.js, Express.js, FastAPI, React, Next.js, REST APIs, WebSockets, WebRTC, Microservices
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, NLP, RAG, Feature Engineering, AutoML
- **Data/Infra:** Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Supabase, Redis, Docker, AWS, Linux, Git, CI/CD

## Experience

**JBN Technologies Pvt. Ltd.** Apr 2026 – Present

*Full Stack Engineering Intern*

- Manage the engineering of CareerGUIDE, an AI-driven career platform spanning frontend development, backend APIs, database architecture, assessment workflows, and AI-assisted career intelligence.
- Built an O*NET ETL pipeline processing 1,016 occupations, 57,543 alternate titles, and 220,098 ratings; developed MBS-based assessment and scoring workflows for personalized career recommendations.

**Analytx4t Lab Pvt. Ltd.** Oct 2025 – Jan 2026

*AI & Backend Engineering Intern*

- Designed and implemented 10+ REST APIs in Node.js for financial analytics workflows spanning backend services, database operations, and application integrations.
- Engineered backend components for data processing, API integration, debugging, and performance optimization in production software.

**ANNAM.AI, IIT Ropar** May 2025 – Jul 2025

*AI/ML Engineering Intern*

- Built end-to-end ML pipelines covering preprocessing, feature engineering, model training, and evaluation using Python, Pandas, and Scikit-learn.
- Trained and evaluated XGBoost and LightGBM models for agricultural intelligence and crop-yield prediction.

## Selected Projects

**Anti-Jailbreak Security System — Pre-LLM Execution Guard** *Python, LLM Security*

- Developed a pre-LLM security layer to detect and mitigate jailbreak attempts, direct and indirect prompt injection, and unauthorized instruction escalation before requests reach downstream AI systems.
- Implemented instruction-hierarchy enforcement, input provenance tracking, pattern-based risk scoring, and multi-turn session analysis to identify adversarial instructions and distinguish trusted directives from untrusted content.
- Designed execution-control routing with ALLOW, BLOCK, and REQUIRE_CONFIRMATION decisions, alongside time-limited capability gating for sensitive operations.
- Repository: github.com/Punyamittal/jailbreak

**AutoML Pipeline — Natural Language to Deployed Model** *Python, ML, Automation*

- Built an automated pipeline that translates natural-language objectives into preprocessing, feature engineering, model selection, hyperparameter tuning, evaluation, and deployment across Random Forest, XGBoost, and LightGBM.

**KAI — Adaptive AI Cognitive Assistant** *Python, LLMs, NLP, Local AI*

- Built an adaptive AI assistant architecture combining conversational context, long-term memory, task extraction, and locally hosted language models for personalized interactions.
- Implemented context-aware memory and emotion-weighted information storage to prioritize relevant past interactions and improve response continuity.
- Experimented with local LLM inference and behavioral feedback mechanisms to enable adaptive, privacy-preserving AI interactions.

## Leadership and Achievements

- **Founder, Youth Season of Code (Y-SoC)** — Built an open-source ecosystem with 250+ members, 100+ projects, and 55+ mentors across 10+ colleges; **Co-Founder, HackFinity**.
- **Secretary, IE(I), VIT Chennai — Technical Lead, VoiceIT Club — 2x Hackathon Winner — NPTEL Top 1%** — Conservation Economics, IIT Kanpur.